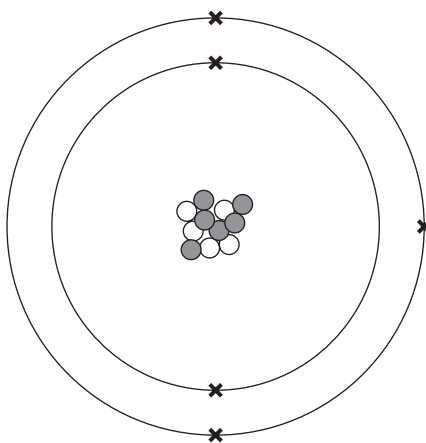


2. (a) Atoms contain particles called electrons, protons and neutrons. The diagram shows an atom of boron.



Complete the sentences.

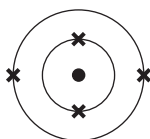
[4]

The nucleus of a boron atom contains five and six

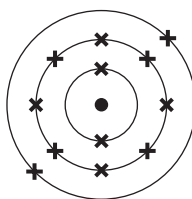
There are five in the shells around the nucleus.

The particles in the atom which have a positive charge are called

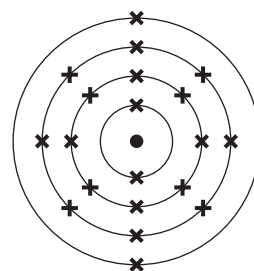
- (b) The diagrams show three different atoms labelled **A**, **B** and **C**. These letters are **not** chemical symbols.



A



B



C

Complete the sentences.

- (i) Atoms **A**, **B** and **C** can be found in Group

[1]

- (ii) The atomic number of atom **A** is

[1]

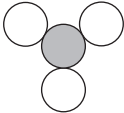
- (iii) The electronic structure of atom **C** is (2,).

[1]

- (iv) Atom **B** can be found in Period

[1]

- (c) The following table shows three substances, their formulae and diagrams that can be used to represent them.

Substance	Formula	Diagram
nitrogen trioxide	NO_3	
methane	CH_4	
water	H_2O	

- (i) Use the information in the table to work out the key being used to represent the different elements in the diagrams. [2]

 is

 is

 is

 is

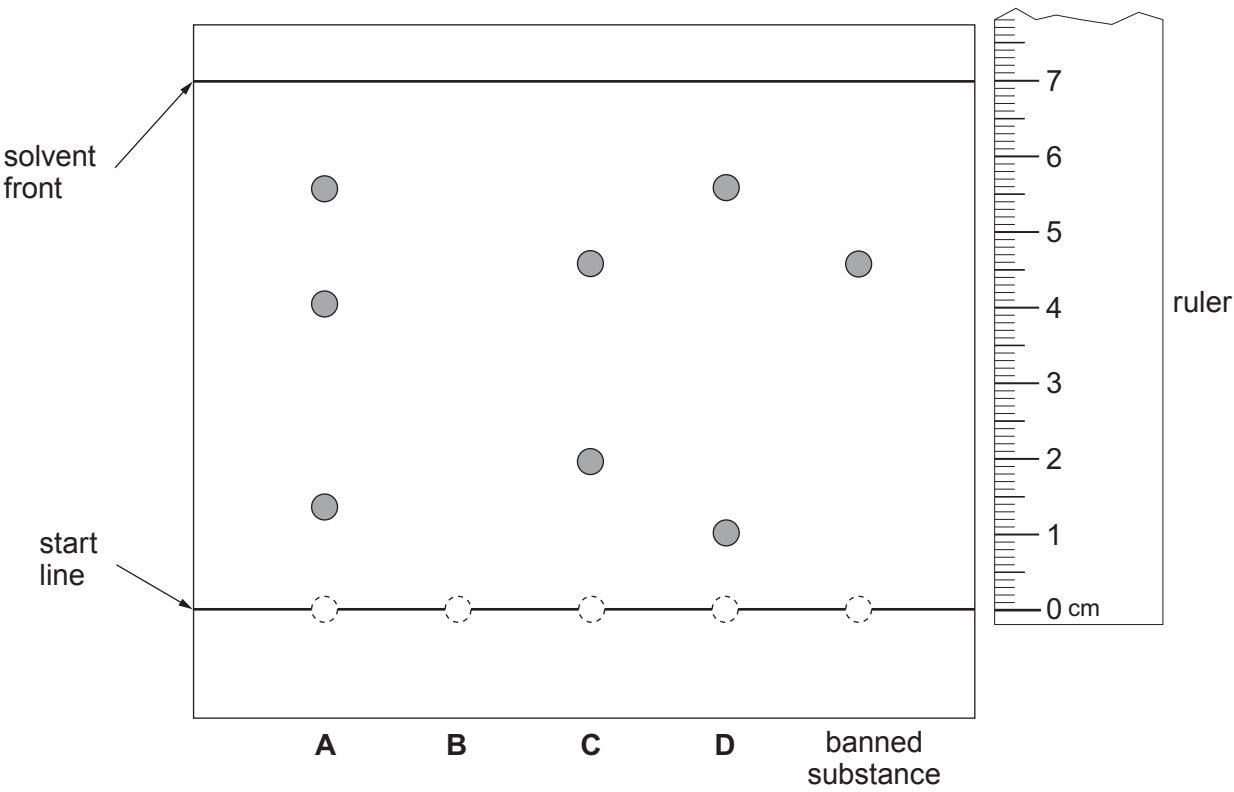
- (ii) **Using the same key** draw a diagram to represent a molecule of carbon dioxide, CO_2 . [1]

- (d) The chemical formula of phosphoric acid is H_3PO_4 .

(i) State how many phosphorus atoms are present in the formula H_3PO_4 [1]

(ii) Give the **total** number of atoms shown in the formula H_3PO_4 [1]

2. A food company was accused of using a banned substance in its sweets. Scientists tested four dyes, **A-D**, to find out if this was true or not. The results are shown below.



(a) Name the method used. [1]

.....

(b) Dye **B** was found to have an R_f value of 0.428.

(i) Use the following equation to calculate the distance moved by dye **B**. [2]

distance moved by dye **B** = $R_f \times$ distance moved by solvent front

Distance moved = cm

(ii) **Complete the diagram** above to show the position of dye **B**. [1]



Examiner
only

(c) Identify which dye, **A-D**, contains the banned substance. Give a reason for your answer. [2]

Dye

Reason

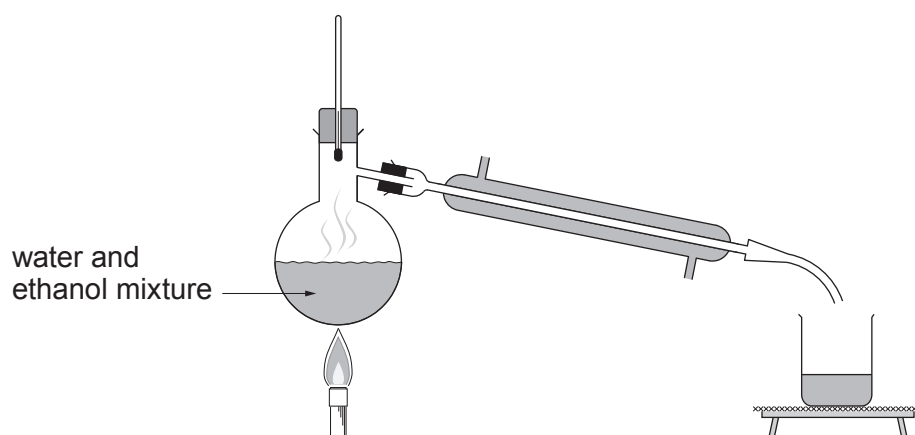
.....

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05



2. (a) A teacher carried out an experiment to show how to separate a mixture of ethanol and water.



- (i) Choose from the box the name of the method used.

[1]

chromatography

filtration

distillation

evaporation

Method

- (ii) The following sentences show the steps involved in this method of separation.

- A** Ethanol vapour cools and condenses
- B** Ethanol boils and evaporates
- C** The mixture of water and ethanol is heated
- D** Ethanol is collected

Put the steps in the correct order. Use the letters **A-D**.

[1]

.....

- (iii) Tick (✓) the box below that explains why it is the water that is left in the flask.

[1]

the boiling point of ethanol is equal to the boiling point of water

☐

the boiling point of ethanol is lower than the boiling point of water

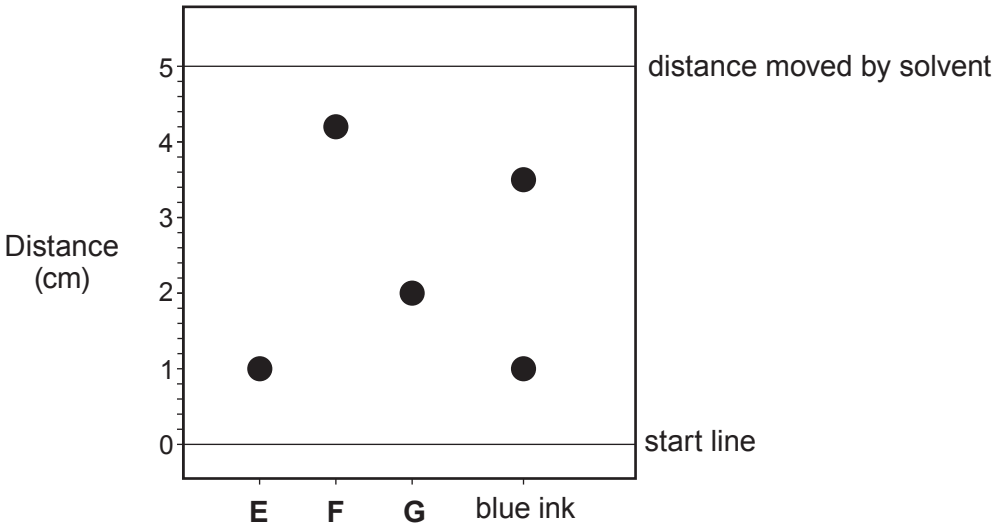
☐

the boiling point of ethanol is higher than the boiling point of water

☐


Examiner
only

(b) A student carried out an investigation to find out whether pigments **E**, **F** and **G** are present in a blue felt pen. The results are shown below.



(i) State what the results tell you about the blue ink. [2]

.....

.....

(ii) Calculate the R_f value of pigment **F** using the equation given. [2]

$$R_f = \frac{\text{distance moved by pigment}}{\text{distance moved by solvent}}$$

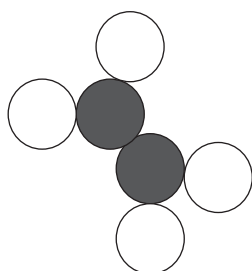
$R_f =$

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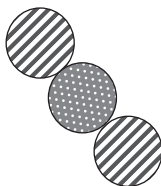
7



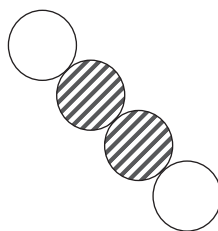
2. (a) The following diagrams represent molecules of some gases.



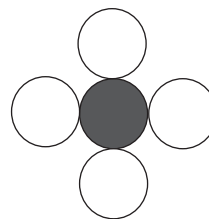
A



B



C



D

Give the **letter** of the diagram which represents each of the following molecules. [2]

sulfur dioxide, SO_2

ethene, C_2H_4

- (b) This chemical equation shows the burning of ethene, C_2H_4 .

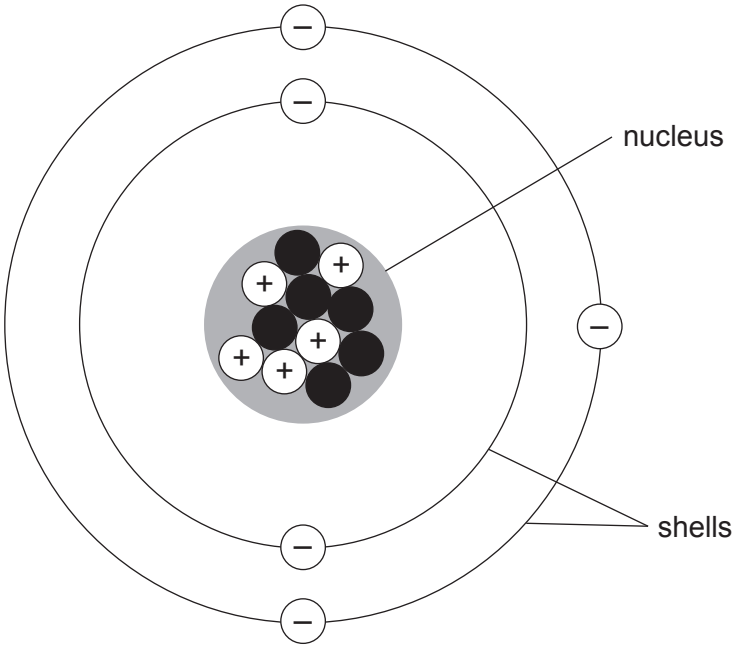


Complete the **word** equation for this reaction. [1]

ethene + oxygen $\longrightarrow$ + water



2. (a) Atoms contain particles called protons, neutrons and electrons. The diagram shows a model of an atom of boron.



State whether the statements in the table below are **true** or **false**. [4]

Statement	True or false?
Boron atoms contain the same number of protons and electrons	
The particles found in the shells are called electrons	
The nucleus contains five neutrons	
The electronic structure of boron is 3,2	



- (b) The formula of boron trioxide is B_2O_3 .

Calculate the relative formula mass (M_r) of boron trioxide.

[2]

$$A_r(\text{B}) = 11 \quad A_r(\text{O}) = 16$$

Relative formula mass =

- (c) The diagrams below represent atoms of boron and fluorine.



boron, B

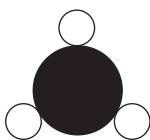


fluorine, F

Boron trifluoride has the formula BF_3 .

Choose the **letter** of the diagram that represents a molecule of boron trifluoride.

[1]



A



B



C

Letter

- (d) Magnesium fluoride contains the ions Mg^{2+} and F^- .

Underline the correct formula for magnesium fluoride.

[1]

